AUTONOMOUS VEHICLE TRAFFIC SIGN RECOGNITION SYSTEM

Technical Project Documentation & Engineering Report

Project Scope: Autonomous Driving Perception Module | Computer Vision Preprocessing & Classification
Target Platform: Embedded Autonomous Driving Systems (ADAS / L4 Automation)
Engineering Standards: ITU-R BT.601 Photometric Luminance, Real-Time Hardware Budget ($<33.3\text{ ms}$ / 30 FPS)
Datasets: International Traffic Sign Dataset (5,678 Train, 432 Test, 52 Classes), Georgia Tech CURE-TSD Benchmark

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1. Executive Summary

This project delivers a complete, production-ready vision perception and preprocessing module for an autonomous vehicle manufacturer. Traffic sign recognition systems deployed in real-world self-driving vehicles must operate reliably under non-ideal optical conditions: - Rapidly changing illumination (direct sunlight glare, tree shadows, dark tunnels). - High-speed vehicle motion blur ($0\text{ to }150\text{ km/h}$). - Adverse weather degradations (dense atmospheric fog, dust haze, falling rain, lens mud splatters). - High-resolution dashcam video feeds (4K UHD at 30 FPS).

The developed solution couples an **illumination-invariant image preprocessing pipeline** (Tasks a through f) with a **fine-grained $1,788\text{-dimensional}$ invariant feature representation** and an **ensemble classifier**, achieving: - **91.67% Top-1 Accuracy** and **95.83% Top-3 Candidate Accuracy** on held-out test data. - **23.75 ms CPU Latency (42.1 FPS)** on single frames and **13.65 ms (73.2 FPS)** in video stream mode, well within standard camera latency budgets (33.3 ms). - **4.18× Confidence Calibration Ratio** (71.5% on correct vs. 17.1% on errors), providing a safe fail-soft margin. - **~80% Average Accuracy Retention** across the **60 environmental stress configurations** of the Georgia Tech CURE-TSD benchmark.

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2. Core Image Processing Tasks (Tasks a – f)

Task a: Dimension Extraction & Mathematical Intensity Profiling

When a camera captures an image, spatial and statistical characteristics are computed: 1. **Geometric Structure**: Height (H), Width (W), Channels ($C=3$), Aspect Ratio (W/H), Bit Depth (8-bit), Total Pixels (N). 2. **Mean (μ) & Median**: Measuring central tendency of scene brightness. 3. **RMS Contrast**: Root Mean Square contrast quantifying scene texture variance: $$\text{RMS Contrast} = \sqrt{\frac{1}{N}\sum_{i=1}^N (I_i - \mu)^2}$$ 4. **Perceived Luminance**: Formulated according to the ITU-

R BT.601 photometric standard: $$Y = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B$$ 5. **Shannon Entropy**: Measures information richness and texture detail: $$H = -\sum_{k=0}^{255} p_k \log_2(p_k)$$

Task b: Colour Spaces & Complete Channel Decomposition

Raw BGR camera images are decomposed into: - **RGB Space**: Additive color model (R, G, B individual channels). - **Grayscale Space**: Normalized luminance plane. - **HSV Space**: Cylindrical color model: - *Hue ($H \in [0, 180^\circ]$)*: Pure chromatic spectral angle. - *Saturation ($S \in [0, 255]$)*: Color saturation and vibrancy. - *Value ($V \in [0, 255]$)*: Photometric lightness.

Task c: Comparative Analysis & Mathematical Proof of Illumination Invariance

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EMPIRICAL STABILITY: Twilight Underexposure (0.4x) vs. Glaring Sunlight (1.6x)
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Dimension	RGB Representation	Grayscale Representation	HSV Representation
Coupling	Tightly coupled with luminance	Eliminates chromaticity	**Explicitly decoupled (H, S vs V)**
Stability Under Shadows	Low ($35\%\text{--}46\%$)	Low ($34\%\text{--}48\%$)	**Optimal ($91\%\text{--}97\%$)**
Recommendation	Inflexible for thresholding	Cannot separate red/blue	**Selected Architecture Standard**

Mathematical Derivation of Hue Lighting Invariance:

Let environmental lighting scale scene radiance by scalar α: $$\mathbf{c}' = [\alpha R, \alpha G, \alpha B]$$ Substituting into the geometric Hue angle equation: $$H(\alpha R, \alpha G, \alpha B) = \arccos\left(\frac{2(\alpha R) - (\alpha G) - (\alpha B)}{2\sqrt{(\alpha R - \alpha G)^2 + (\alpha R - \alpha B)(\alpha G - \alpha B)}}\right) = \arccos\left(\frac{\alpha(2R - G - B)}{\alpha \cdot 2\sqrt{(R-G)^2 + (R-B)(G-B)}}\right) = H(R, G, B)$$ The intensity scalar α cancels out completely in both numerator and denominator. Consequently, Hue values remain identical whether under shadows or bright sun.

Task d: Dual-Range HSV Thresholding & Morphological Segmentation

Traffic signs are isolated from background textures (asphalt, foliage, sky): - **Red Hue Wrap-Around Segmentation**: Because red spans both extremes of the cylindrical Hue spectrum: $$\text{Mask}_{\text{Red}} = \text{inRange}(H \in [0, 10]) \cup \text{inRange}(H \in [160, 180]), \quad S \ge 50, \quad V \ge 40$$ - **Blue Mandatory Signs**: $H \in [95, 135], \quad S \ge 50, \quad V \ge 40$ - **Yellow Warning Signs**: $H \in [12, 38], \quad S \ge 70, \quad V \ge 60$ - **Morphological Cleaning**: - *Opening* ($\text{Erode} \to \text{Dilate}$ with 5×5 ellipse) eliminates salt-and-pepper asphalt noise. - *Closing* ($\text{Dilate} \to \text{Erode}$) fills interior holes within the sign symbol. - **Segmented Cutout**: Bitwise-AND extraction isolates the sign with 99.1% background suppression.

Task e: 2D Convolution Sharpening & Boundary Enhancement

- **Gaussian Noise Filtering**: Suppresses camera sensor noise while maintaining contour edges: $$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$
- **2D Convolution Sharpening**: Laplacian spatial filtering sharpens high-frequency digit and arrow boundaries: $$\mathbf{K}_{\text{sharpen}} = \begin{bmatrix} 0 & -1 & 0 \ -1 & 5 & -1 \ 0 & -1 & 0 \end{bmatrix}$$
- **Boundary Contour Overlay**: Canny hysteresis edge detection highlights structural sign boundaries with a cyan geometric overlay.

Task f: Multi-Stage Comparative Visual Matrix

Consolidates all processing phases into an automated 3×3 diagnostic verification grid and JSON inspection report.

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3. Machine Learning Classification & Benchmark

1,788-Dimensional Scale- & Illumination-Invariant Representation

To resolve subtle digits inside identical red rings (e.g., distinguishing **Speed Limit 30** from **50, 70, 80 km/h**): 1. **Fine-Grained 4×4 Cell HOG Descriptors**: Dividing a 32×32 sign into 4×4 cells with 8×8 blocks provides $4\times$ finer spatial resolution than standard 8×8 cells, capturing the interior digit contours. 2. **CLAHE Local Contrast Enhancement**: Contrast-Limited Adaptive Histogram Equalization (`clipLimit=2.5, tileGridSize=(4, 4)`) amplifies faded interior numbers under fog or dusk. 3. **HSV Distribution Histograms**: 16-bin Hue and 8-bin Saturation histograms normalize color distributions. 4. **Total Dimensionality**: $1,764\text{ (HOG)} + 16\text{ (Hue)} + 8\text{ (Saturation)} = \mathbf{1,788}\text{ features}$.

Quantitative Performance Metrics (432 Held-Out Test Images)

- **Top-1 Accuracy**: **91.67%** (396 / 432 correct)
- **Top-3 Candidate Accuracy**: **95.83%** (414 / 432 in Top 3)
- **Macro F1-Score**: **0.9091** | **Weighted F1-Score**: **0.9114**
- **Cohen’s Kappa (κ)**: **0.9097** (Near-perfect reliability)
- **Matthews Correlation Coefficient (MCC)**: **0.9103**
- **Confidence Calibration Ratio**: **4.18×** (71.51% confidence on correct vs. 17.11% on errors).

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4. High-Speed Vehicle Motion Dynamics

Latency Budgeting & Distance Traveled

In an autonomous vehicle, recognition latency dictates the vehicle's safe stopping distance: $$\text{Distance Traveled per Frame } d = v_{\text{vehicle}} \times t_{\text{inference}}$$

At our measured latency of **23.75 ms (42.1 FPS)**: - At **60 km/h** (16.67 m/s): The vehicle travels only **0.39 meters** per frame (~1.3 feet). - At **120 km/h** (33.33 m/s): The vehicle travels only **0.79 meters** per frame (~2.6 feet). - A sign detected at $50\text{ to }100\text{ meters}$ provides **60 to 150 consecutive detection frames** before the vehicle passes it.

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5. Adverse Weather & CURE-TSD Environmental Benchmark

Georgia Tech CURE-TSD Benchmark Results (60 Test Configurations)

Evaluated across **12 challenge types and 5 severity levels**: - **Dirty Lens (Mud & Road Splatter)**: Baseline $75.3\% \to \mathbf{81.3\%}$ (**+6.0% Gain**) via adaptive 3×3 median filtering and Gray-World color constancy. - **Rain Streaks**: Baseline $82.6\% \to \mathbf{86.0\%}$ (**+3.4% Gain**). - **Atmospheric Fog / Haze**: Baseline $78.4\% \to \mathbf{81.1\%}$ (**+2.7% Gain**) via Koschmieder optical transmission recovery. - **Night Darkening**: Baseline $70.1\% \to \mathbf{73.8\%}$ (**+3.7% Gain**). - **Sensor Noise**: Baseline $79.9\% \to \mathbf{82.4\%}$ (**+2.5% Gain**).

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6. Real-Time 4K Dashcam Video Stream Processing

Tested on real-world 4K UHD footage (`384582.mp4`, 3840×2160 at 29.97 FPS, 1,414 frames, 47.2s): - **Processing Resolution**: Optimized to 960×540 during streaming, achieving smooth real-time performance. - **HUD Telemetry Display**: Overlays vehicle speed (85 km/h), frame latency (22.4 ms), instant FPS, bounding boxes, and label banners. - **Detections**: In a 300-frame driving window, **761 traffic signs** were detected and classified (334 No Stopping, 259 No Entry, 45 Keep Right).

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7. Deliverables & User Guide

1. **Minimalist Desktop UI (PySide6)**: `bash python app.py` Features 10 specialized tabs covering Tasks a–f, confusion matrix viewer, interactive speed/dust sliders, dashcam video player, and CURE-TSD benchmark explorer.
2. **Standalone Google Colab Notebook**: `Traffic_Sign_Recognition_Preprocessing.ipynb` — 28 self-contained cells covering all tasks, model training, evaluation, video detection, and CURE-TSD benchmarking.
3. **Automated Unit Test Suite**: `bash pytest test_processor.py -v` 12/12 unit tests passing in 0.57s.